

Problem Statement:

Azure Firewall Manager Policy Hierarchy Design

Team Members:

1.GUDAPATI SURENDRA - 2400031555

2. GUDIVAKA ISHWARYA - 2400031550

3. JONNADULA YASASWINI - 2400090025

4. KONDEPATI VISWANATH - 2400090021

Mentor: Dr. Sandeep Kumar Sharma

What is a Firewall?

- A firewall controls network traffic between different networks.
- It checks traffic based on predefined rules.
- It can allow or block traffic.
- It helps protect applications and resources from unwanted network access.

Azure Firewall

- Azure Firewall is a managed firewall service provided by Microsoft Azure.
- It can control network traffic using firewall policies and rules.

Problem Statement

When an organization has multiple applications or environments, maintaining separate firewall rules can become difficult.

For example:
Development may need different rules.
Finance may need different rules.
Some rules are common for all applications.
Managing all these rules separately can become difficult.

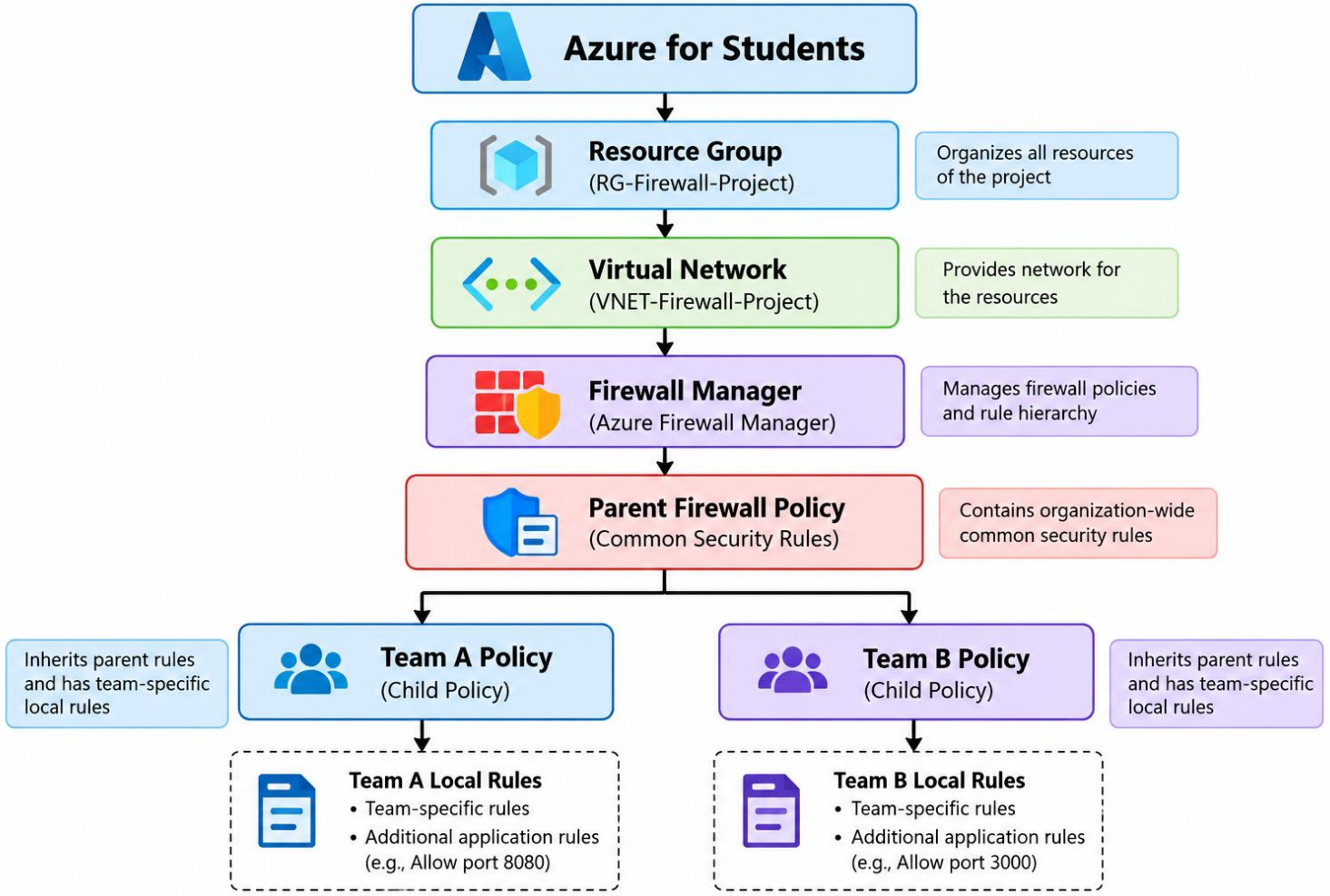
Our Solution:
We use Azure Firewall Manager Policy Hierarchy where:
Parent Policy → Common Rules
Child Policies → Specific Rules

This allows common rules to be inherited while adding application-specific rules.

Objectives

- **Centralized Security**
  To centrally manage common firewall security rules using Azure Firewall Manager.
- **Policy Hierarchy**
  To design a parent-child firewall policy hierarchy where child policies inherit organization-level rules.
- **Controlled Customization**
  To allow individual teams or applications to define their own additional firewall rules while maintaining the organization's common security requirements.
- **Simplified Management**
  To reduce duplicate rule configuration and provide consistent security management across multiple Azure environments.

Project Flow



Policy Hierarchy

Policy Purpose Rule:
fw-parent-policy: Common policy [TCP 80, 443]
fw-child-policy Application 1: Inherits 80,443 + TCP 8080
fw-child-policy-2 Application 2: Inherits 80,443 + TCP 9090

Example:

Parent: Allow Web Traffic → 80, 443

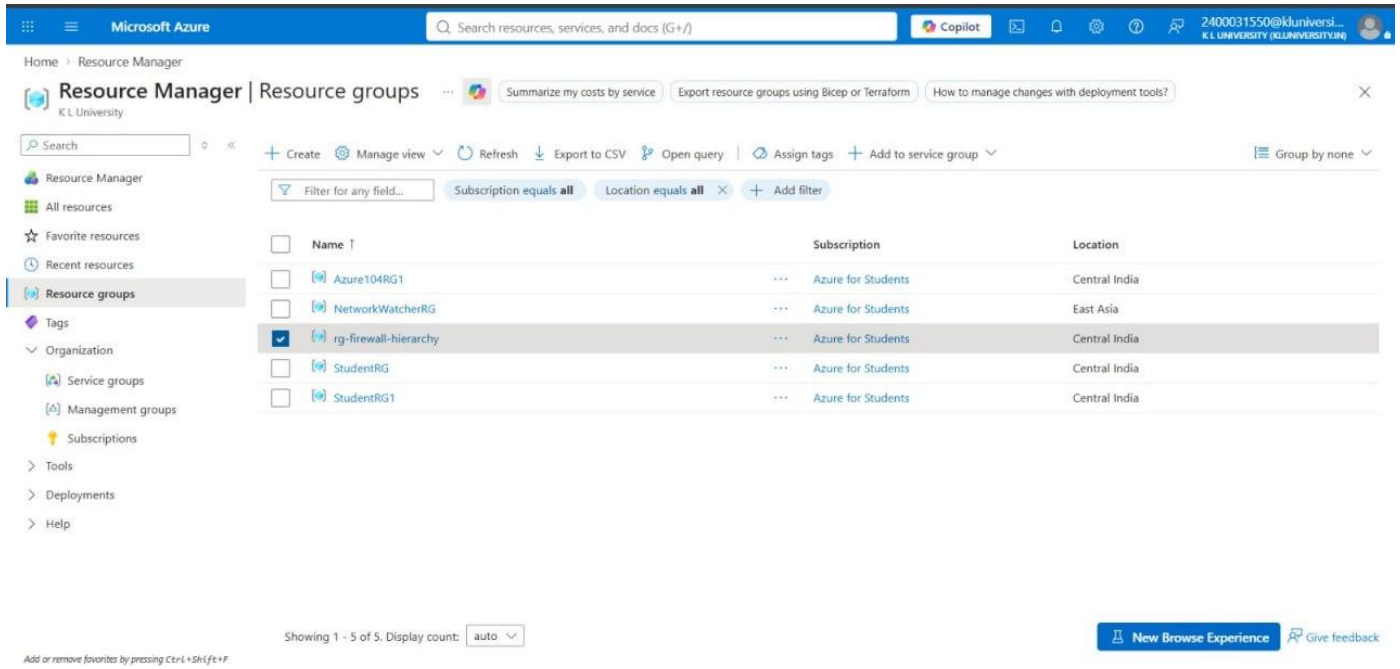
Child 1: Adds Application Traffic → 8080

Child 2: Adds Application Traffic → 9090

This shows how different applications can have common and specific rules.

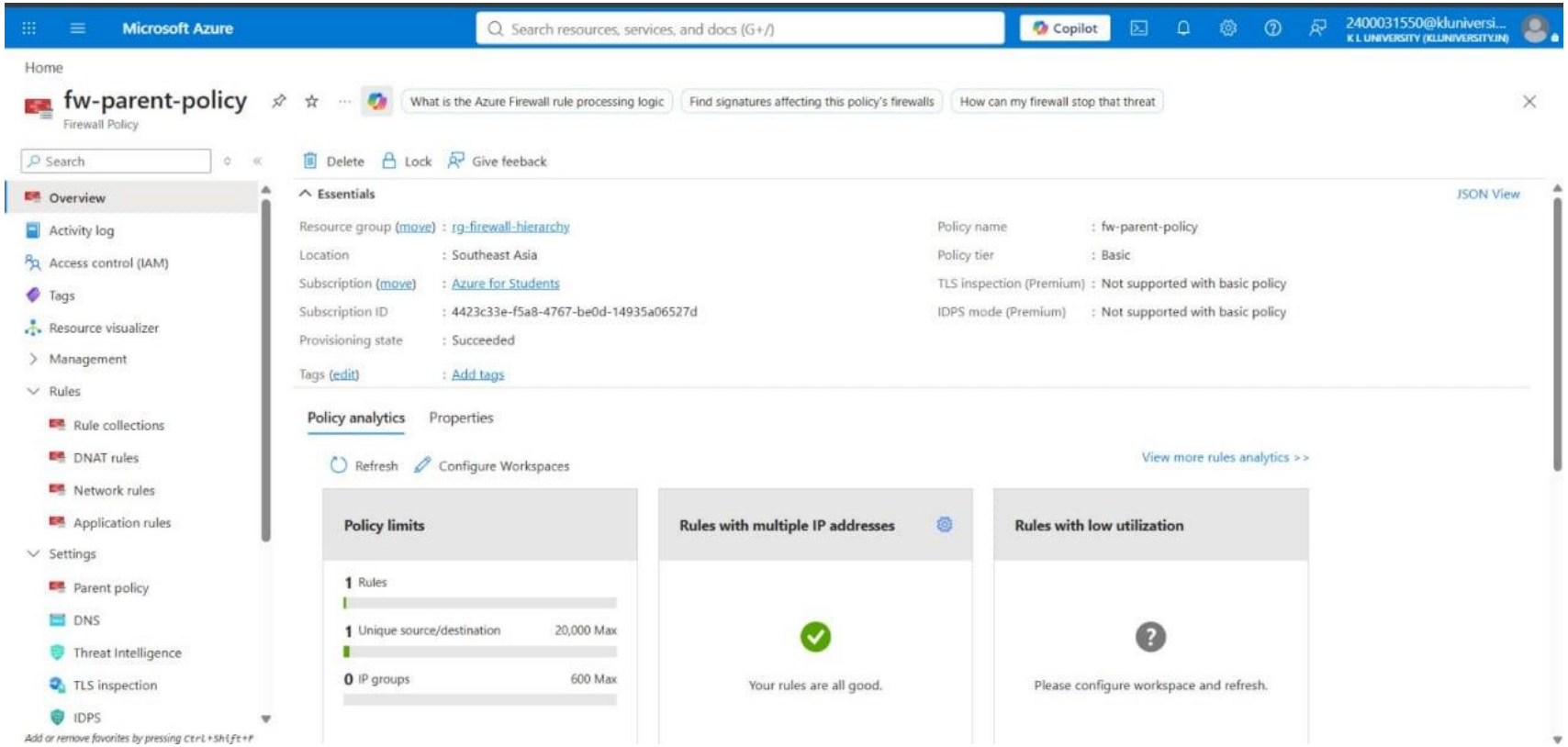
Step 1 — Create Resource Group

We first created a separate resource group to keep all project resources together.



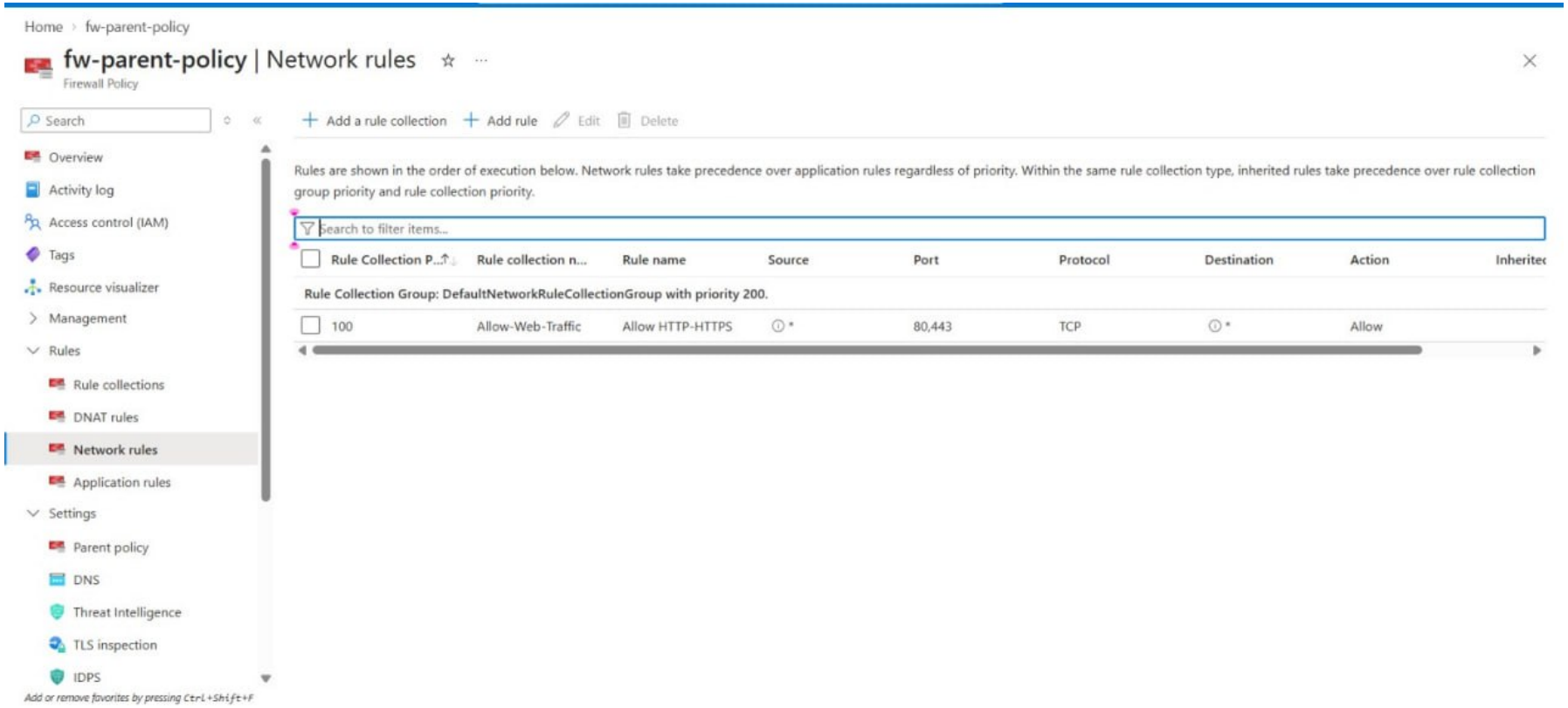
Step 2 — Create Parent Firewall Policy

This is our **main/base policy**.

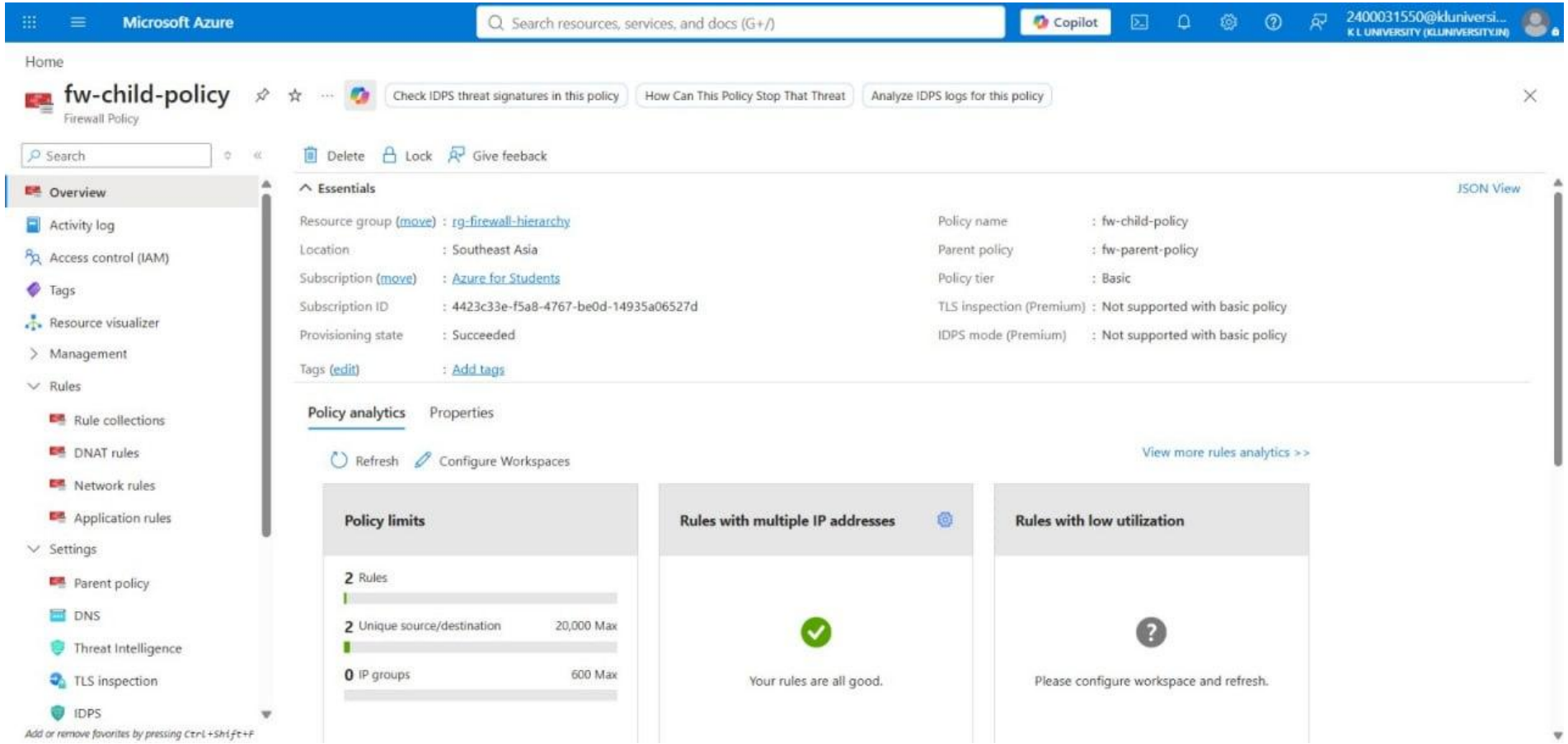


Step 3 — Add Common Network Rule

Inside fw-parent-policy, we created a network rule collection.

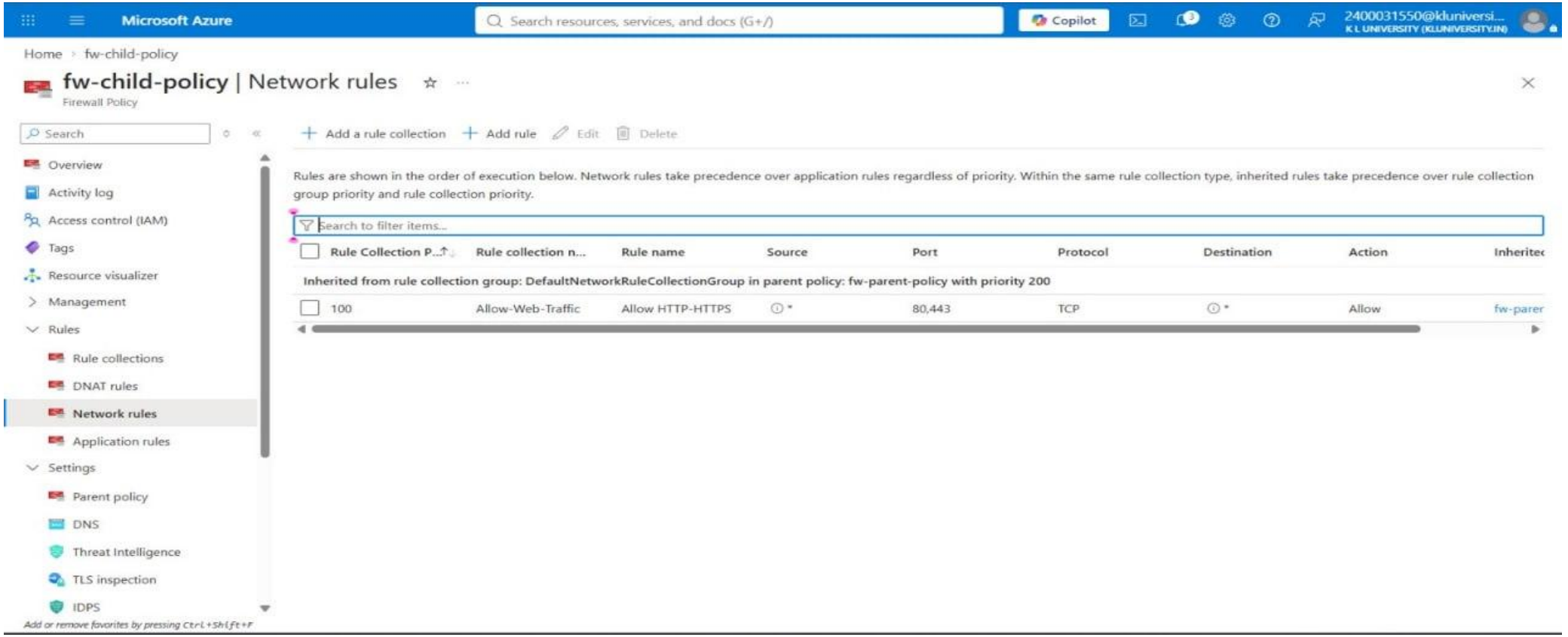


Step 4 — Create Child Firewall Policy



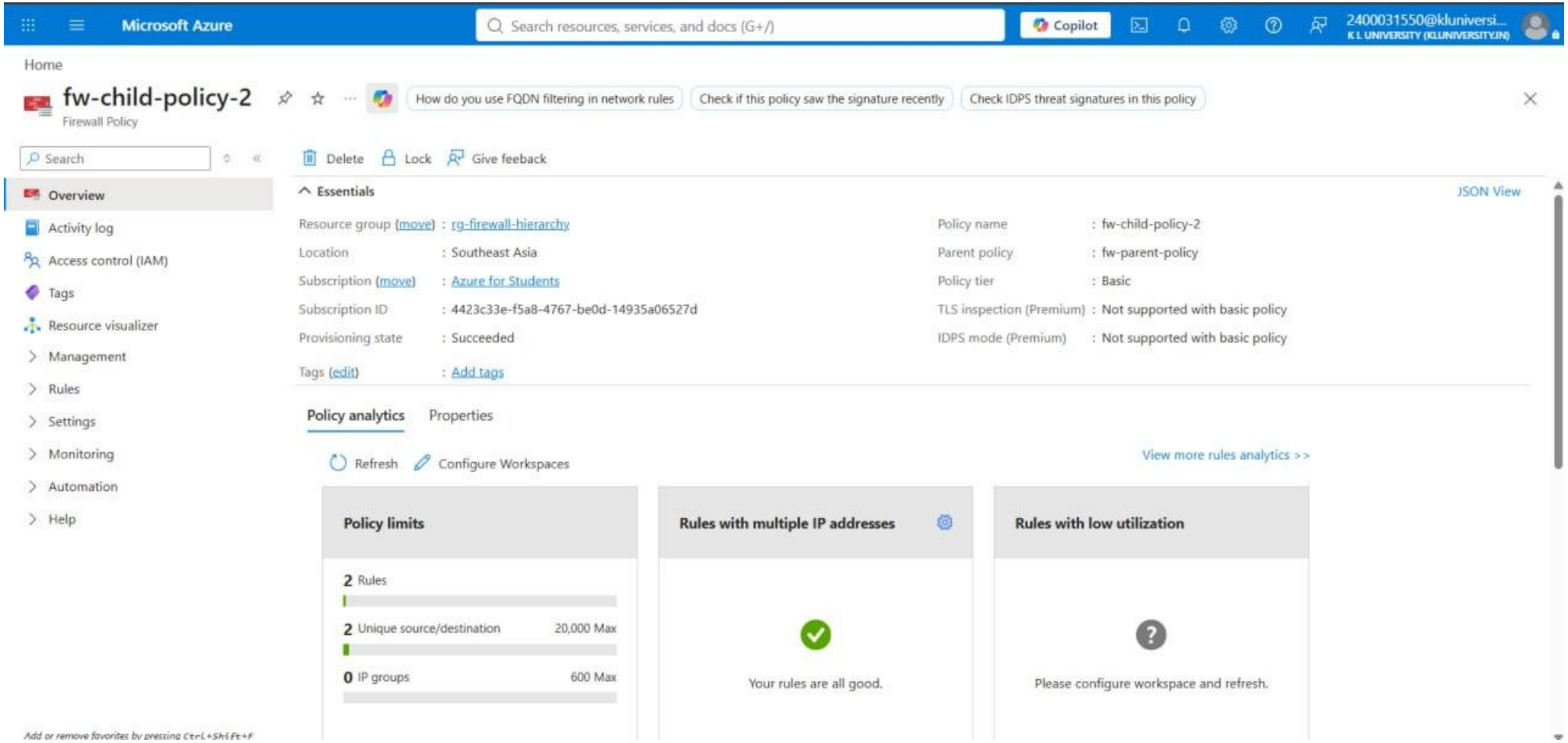
Step 5 — Add Child-Specific Rule

Inside fw-child-policy, we created another network rule collection.



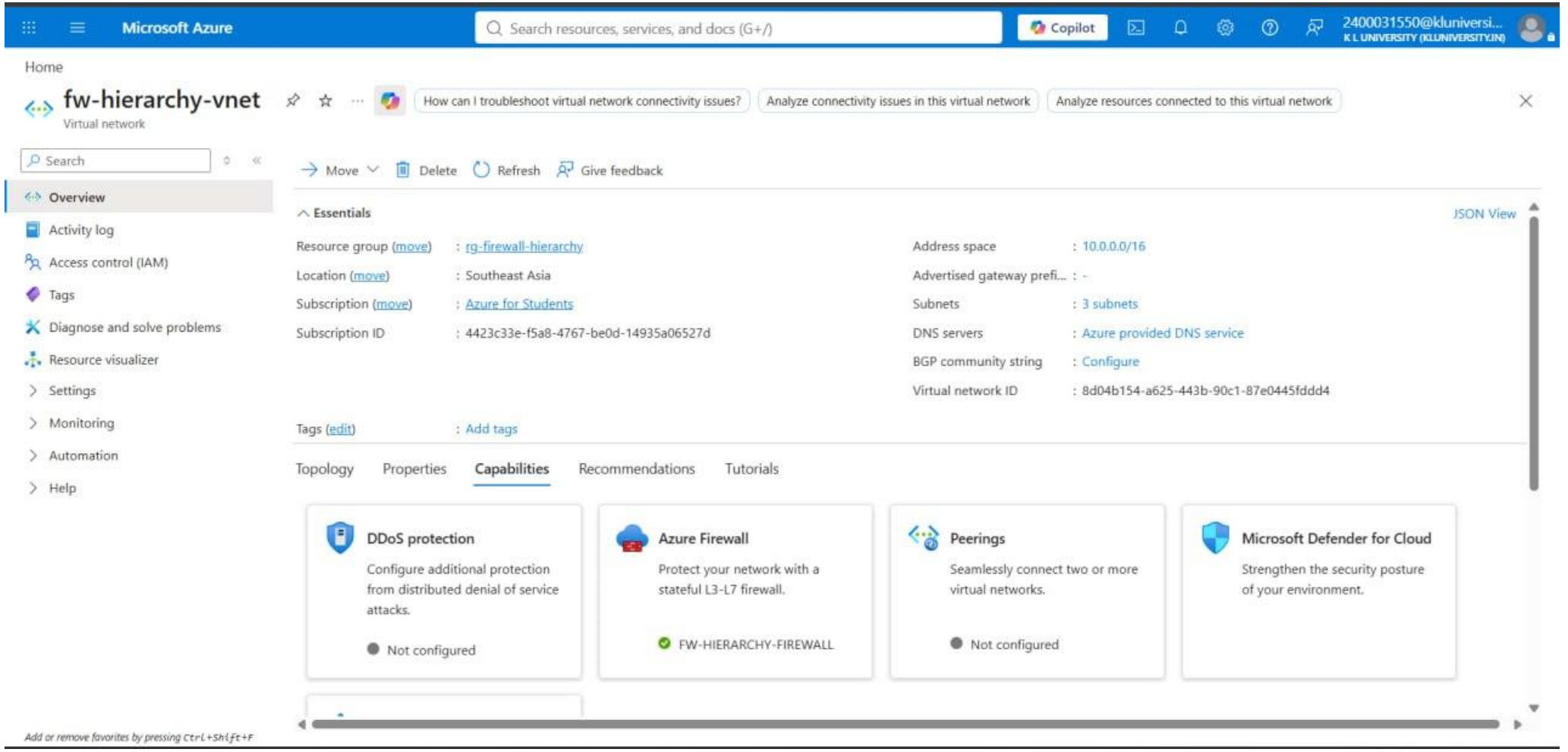
Step 6 — Create Second Child Policy

To demonstrate that different environments can have different rules, we created another child policy.



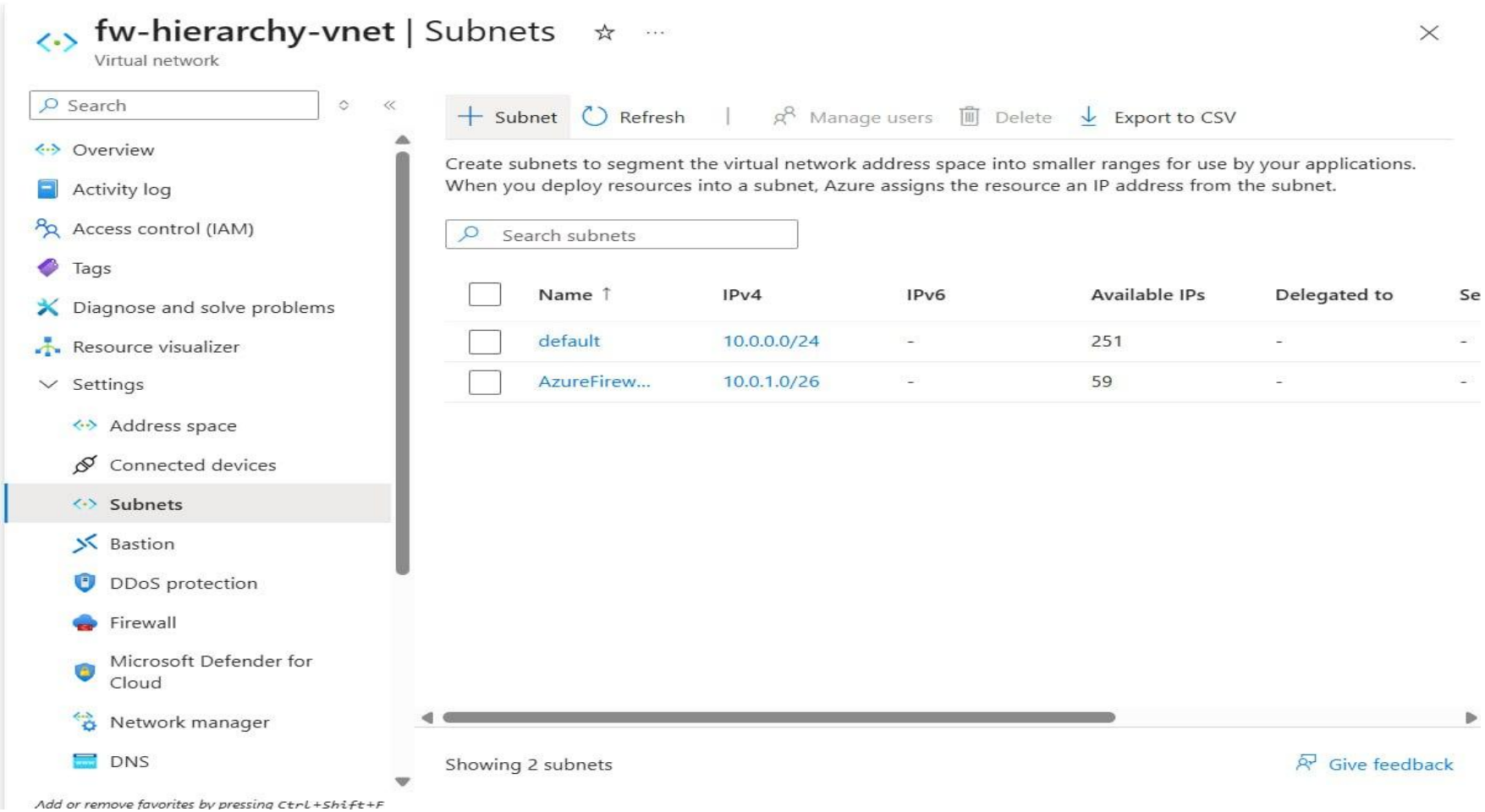
Step 7 — Create Virtual Network

This provides the network environment for our firewall and test VM.

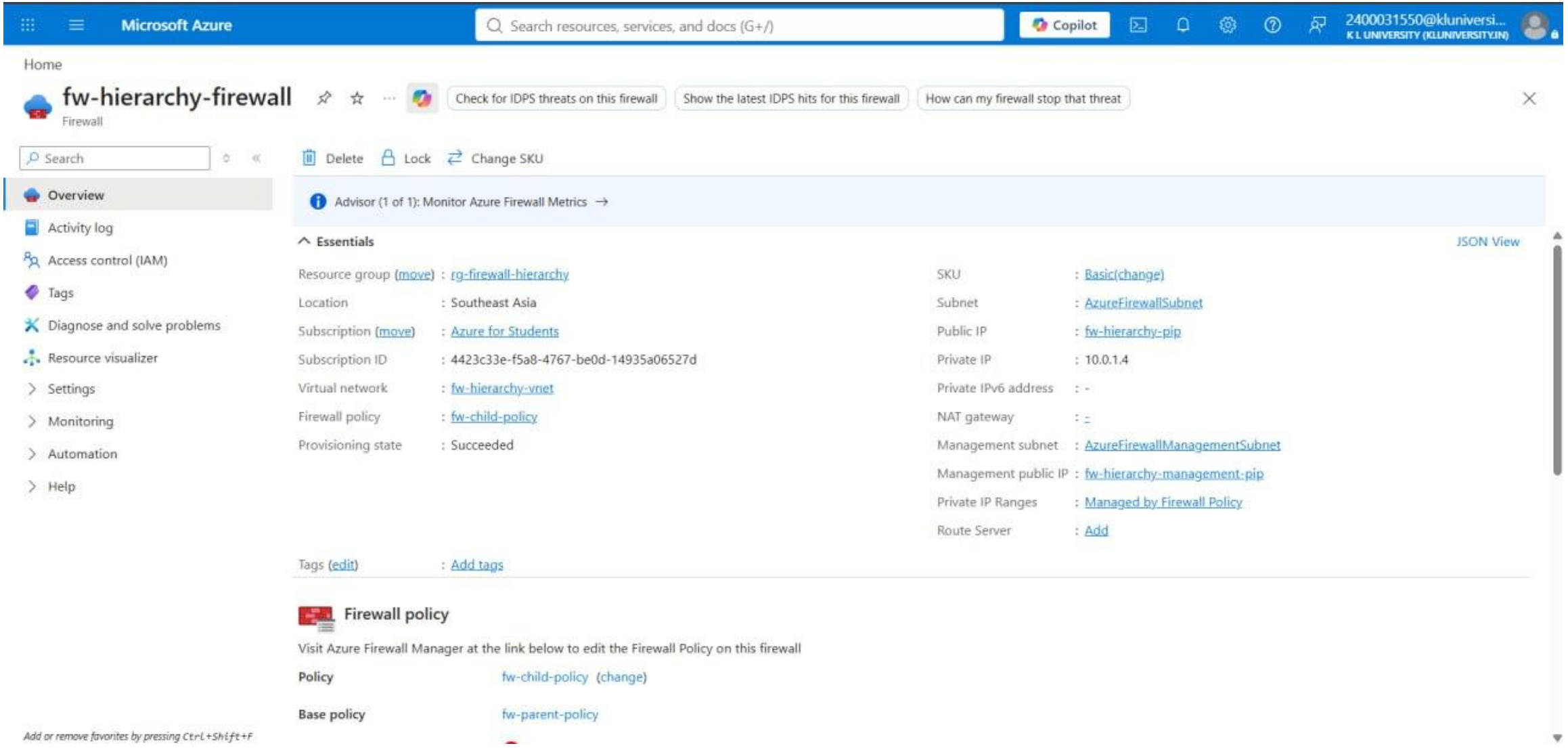


Step 8 — Create Required Subnets

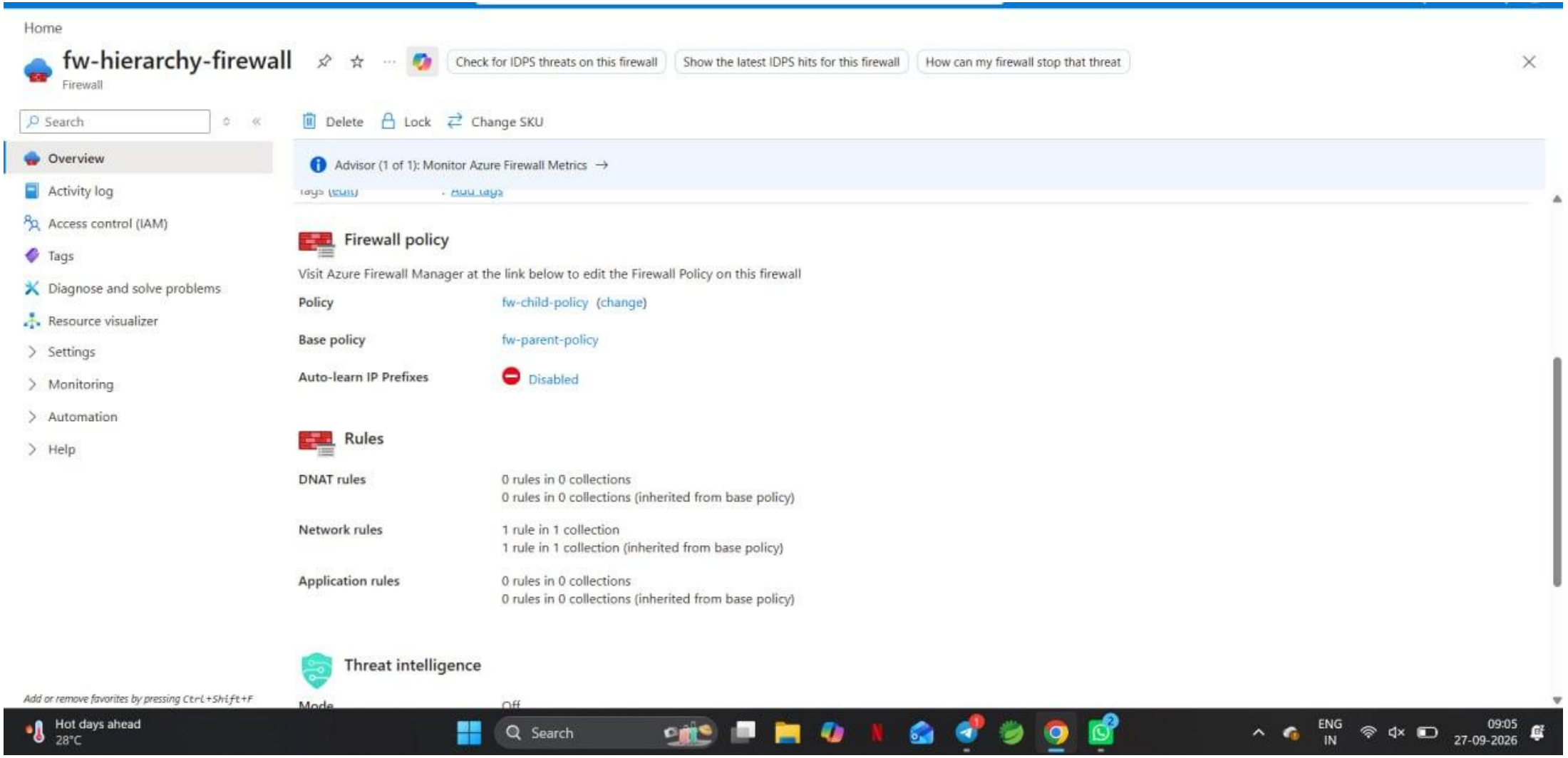
Used for our test VM, Azure Firewall, firewall management



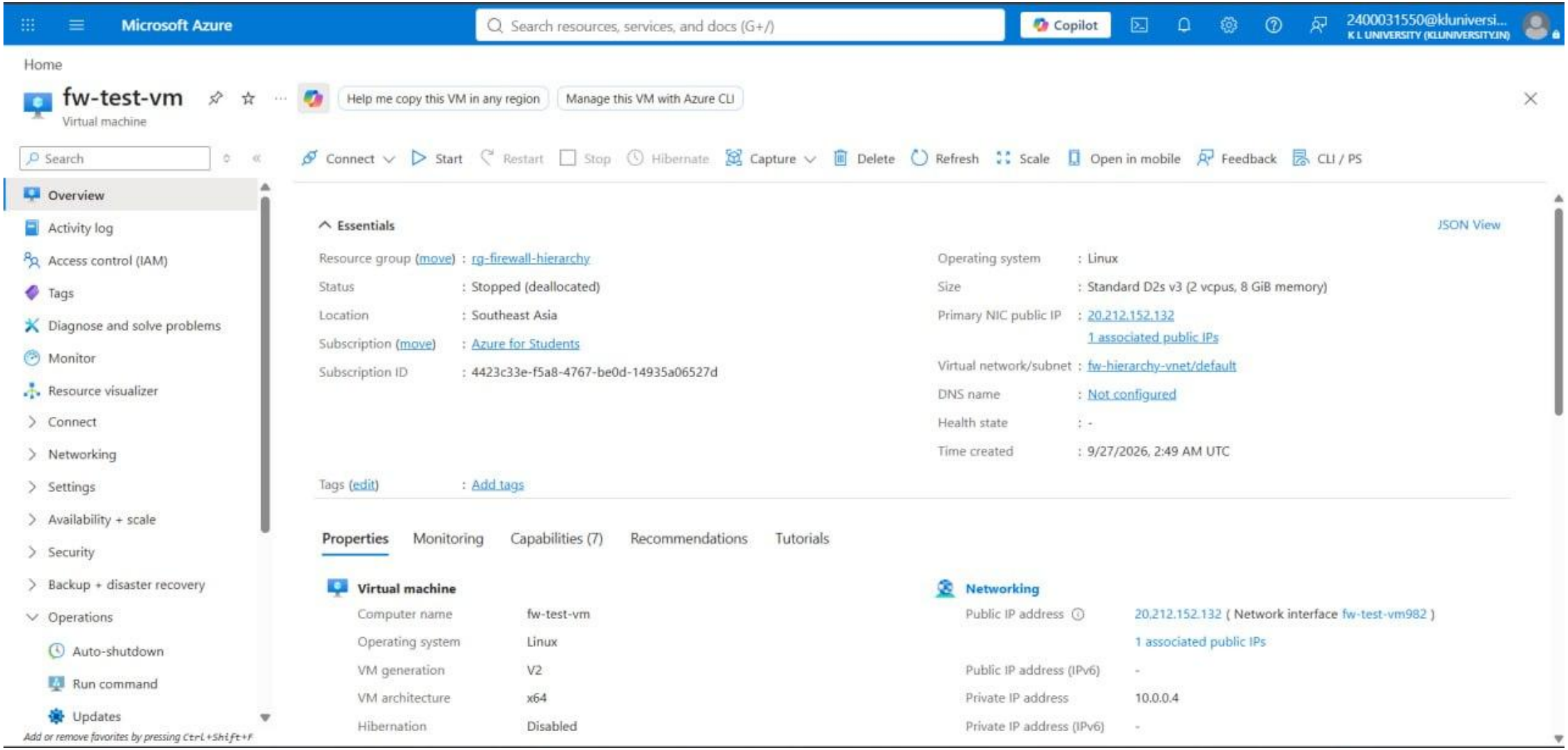
Step 9 — Deploy Azure Firewall



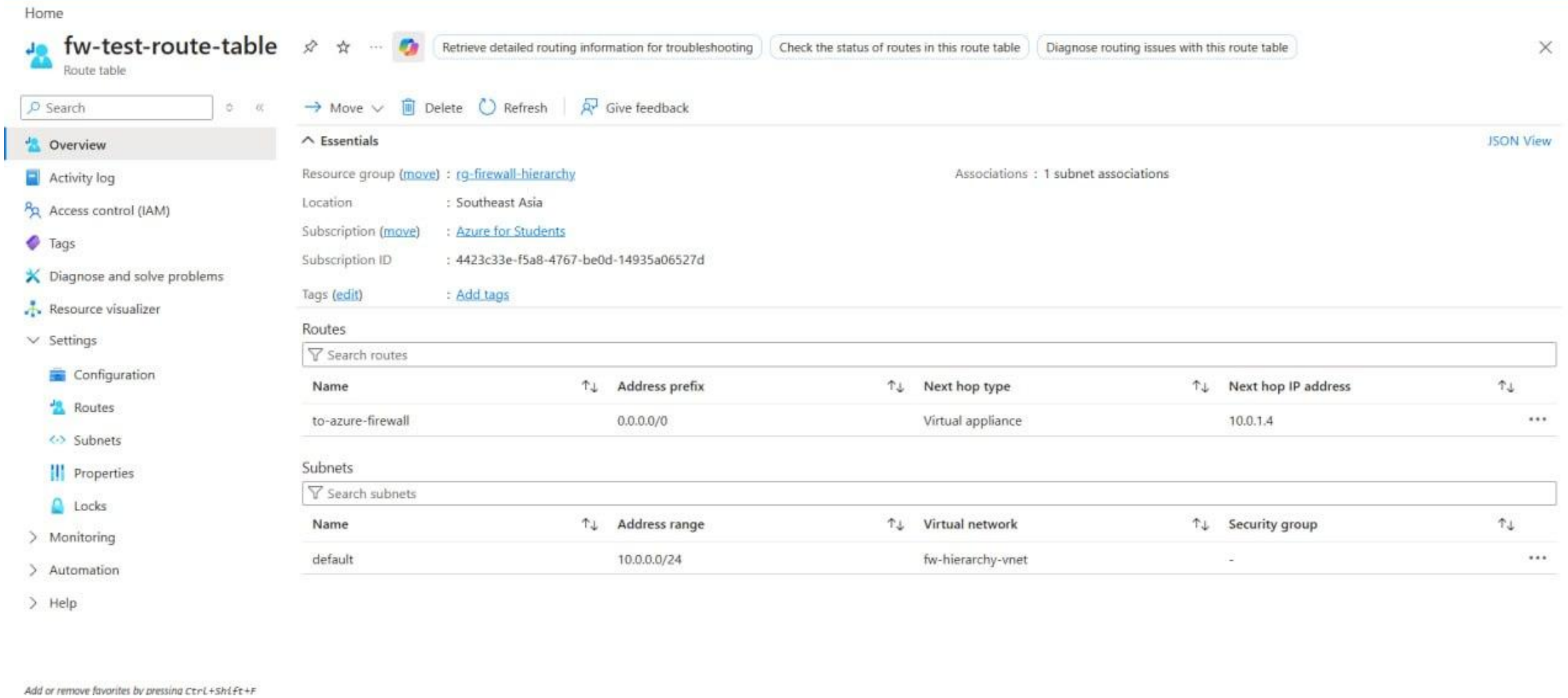
Step 10 — Verify Policy Hierarchy



Step 11 — Create Test VM



Step 12 — Create Route Table



Step 13 — Add Default Route

Routes

Search routes

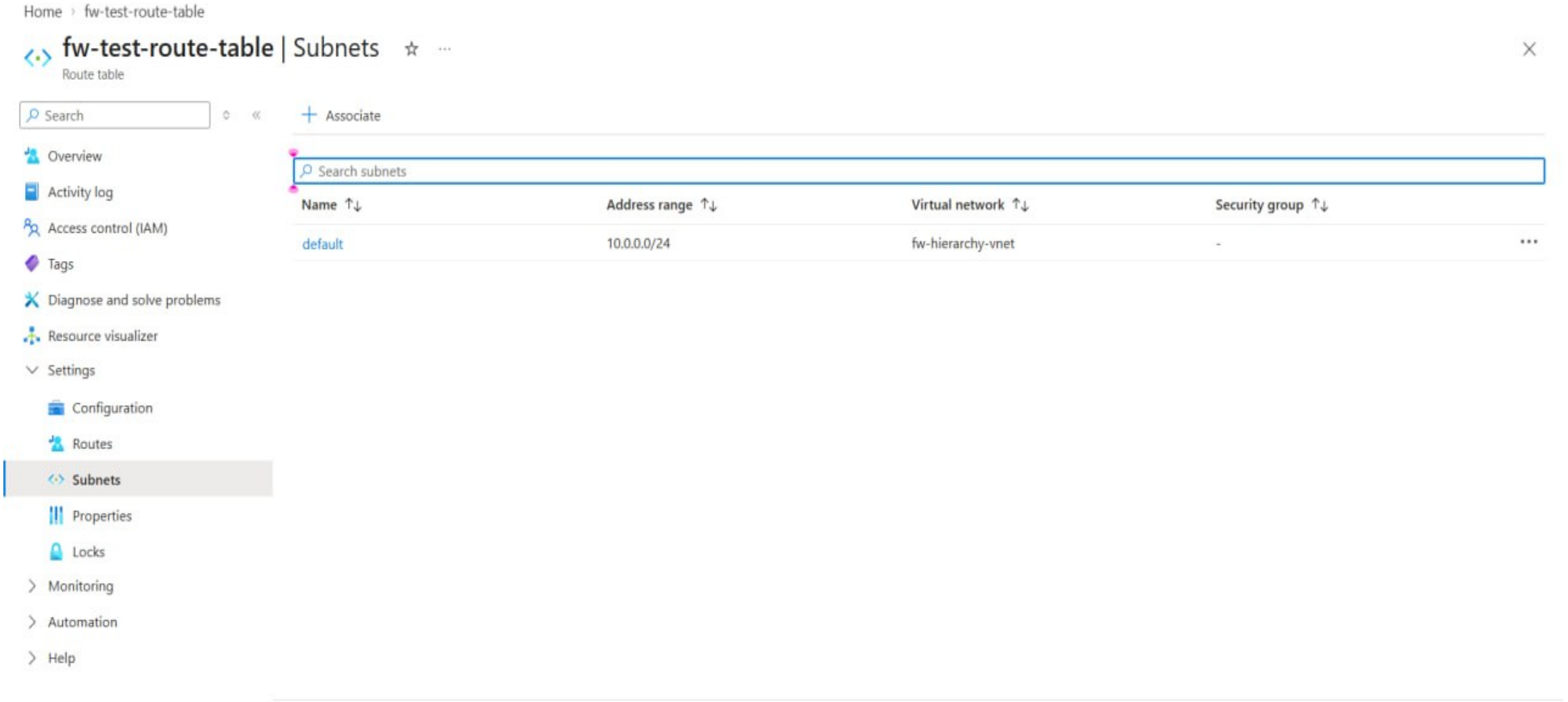
Name	Address prefix	Next hop type	Next hop IP address	
to-azure-firewall	0.0.0.0/0	Virtual appliance	10.0.1.4	•••

Subnets

Search subnets

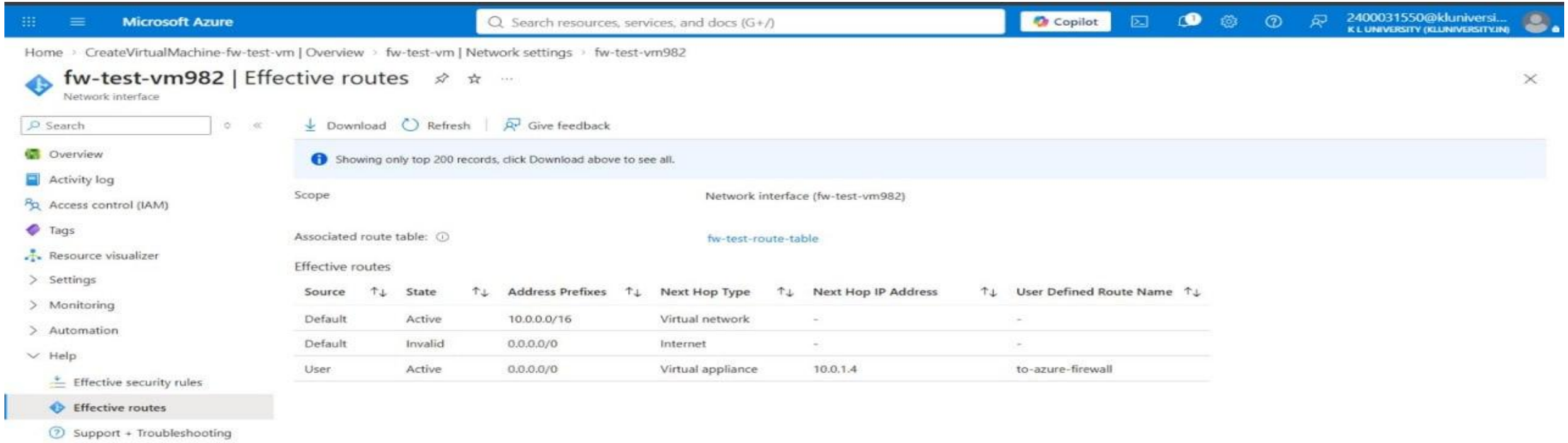
Name	Address range	Virtual network	Security group	
default	10.0.0.0/24	fw-hierarchy-vnet	-	•••

Step 14 — Associate Route Table with VM Subnet

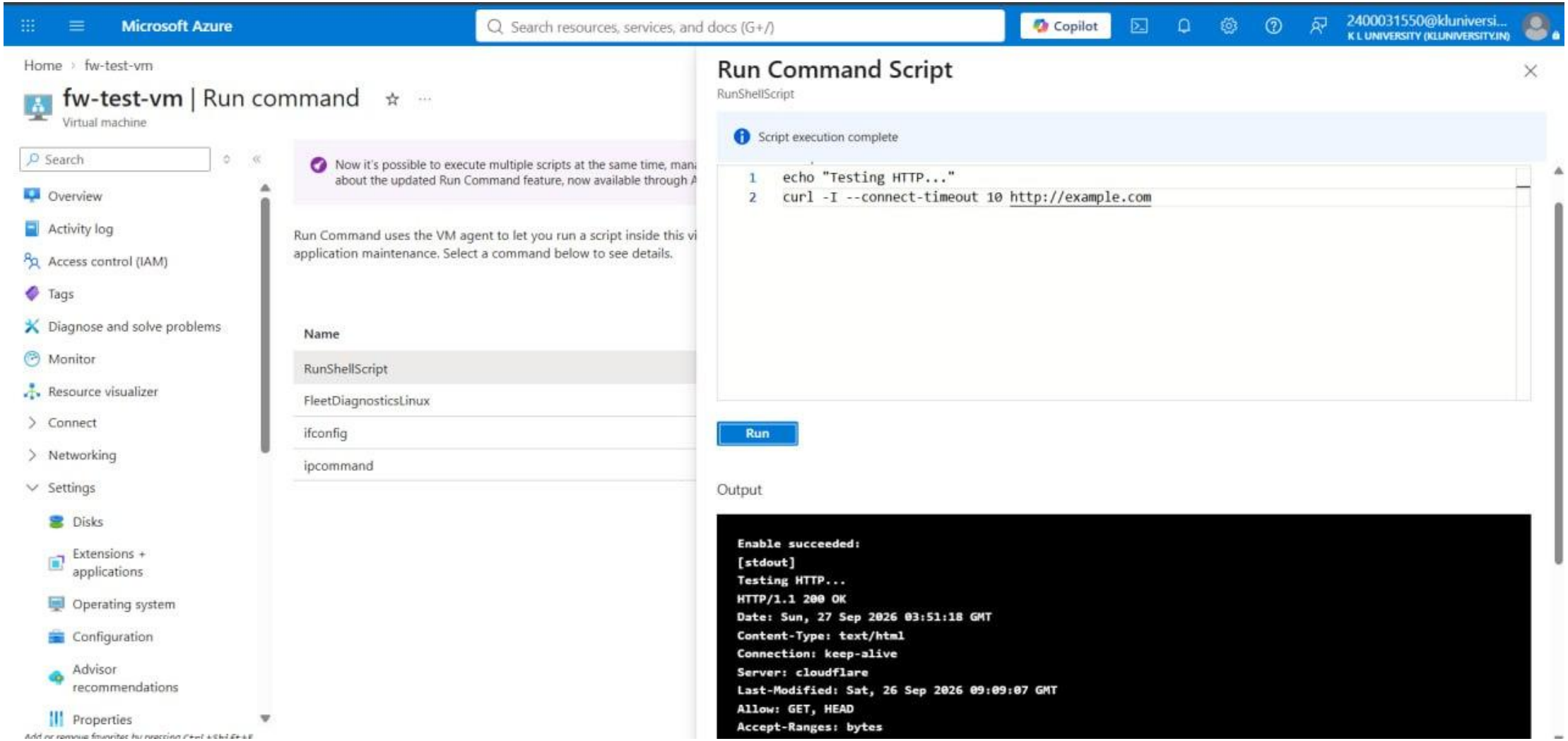


Step 15 — Verify Effective Route

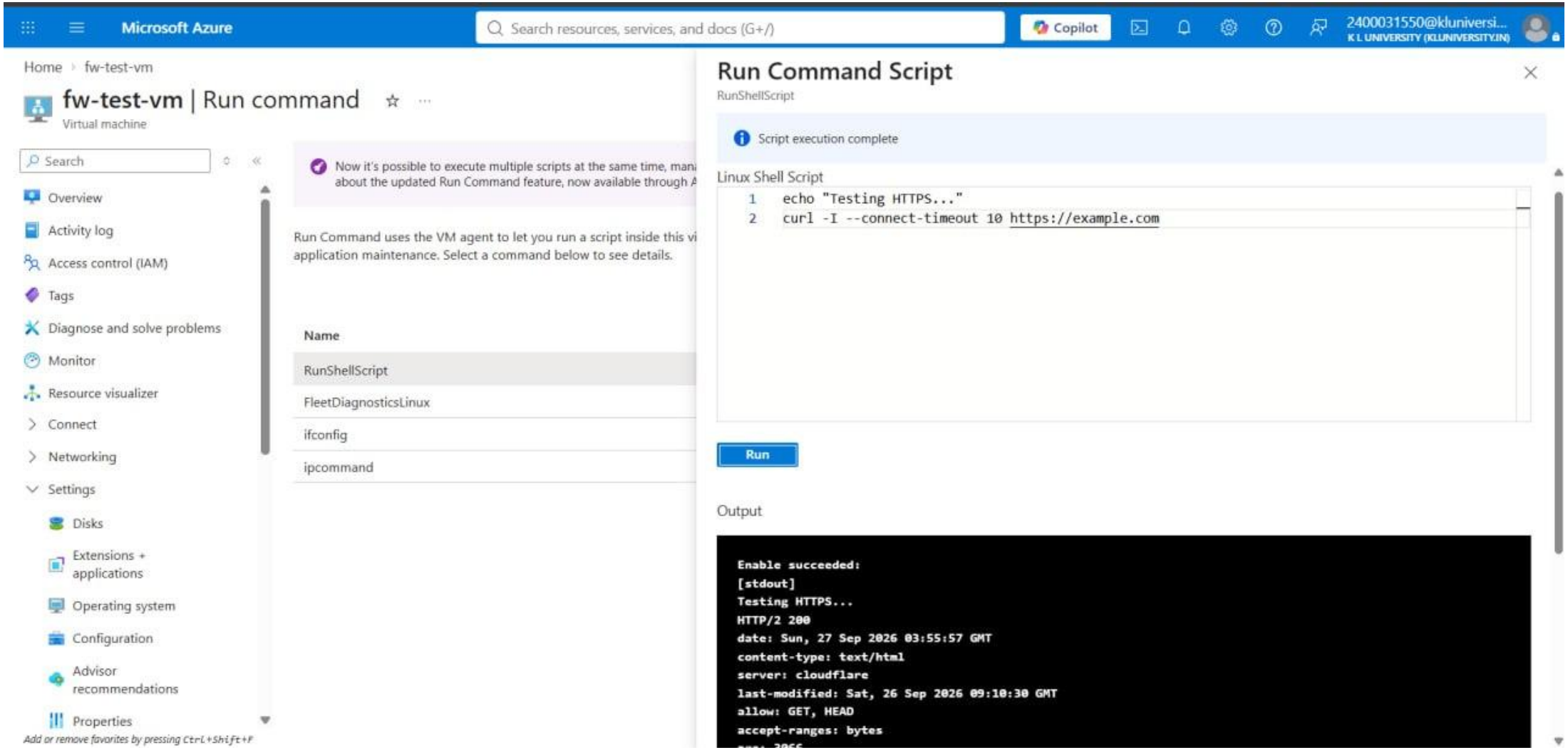
This was one of our most important verification steps.



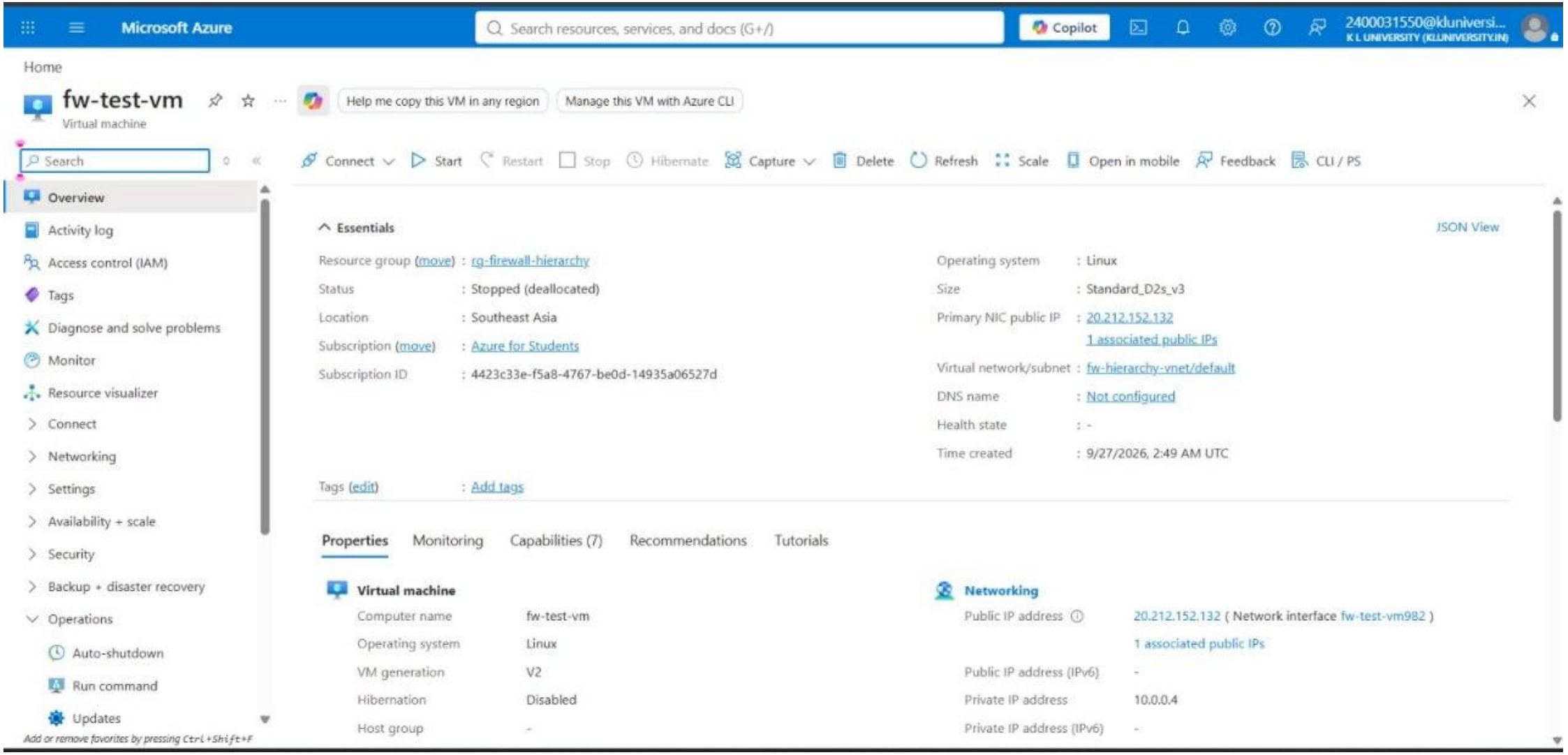
Step 16 — Test HTTP Traffic



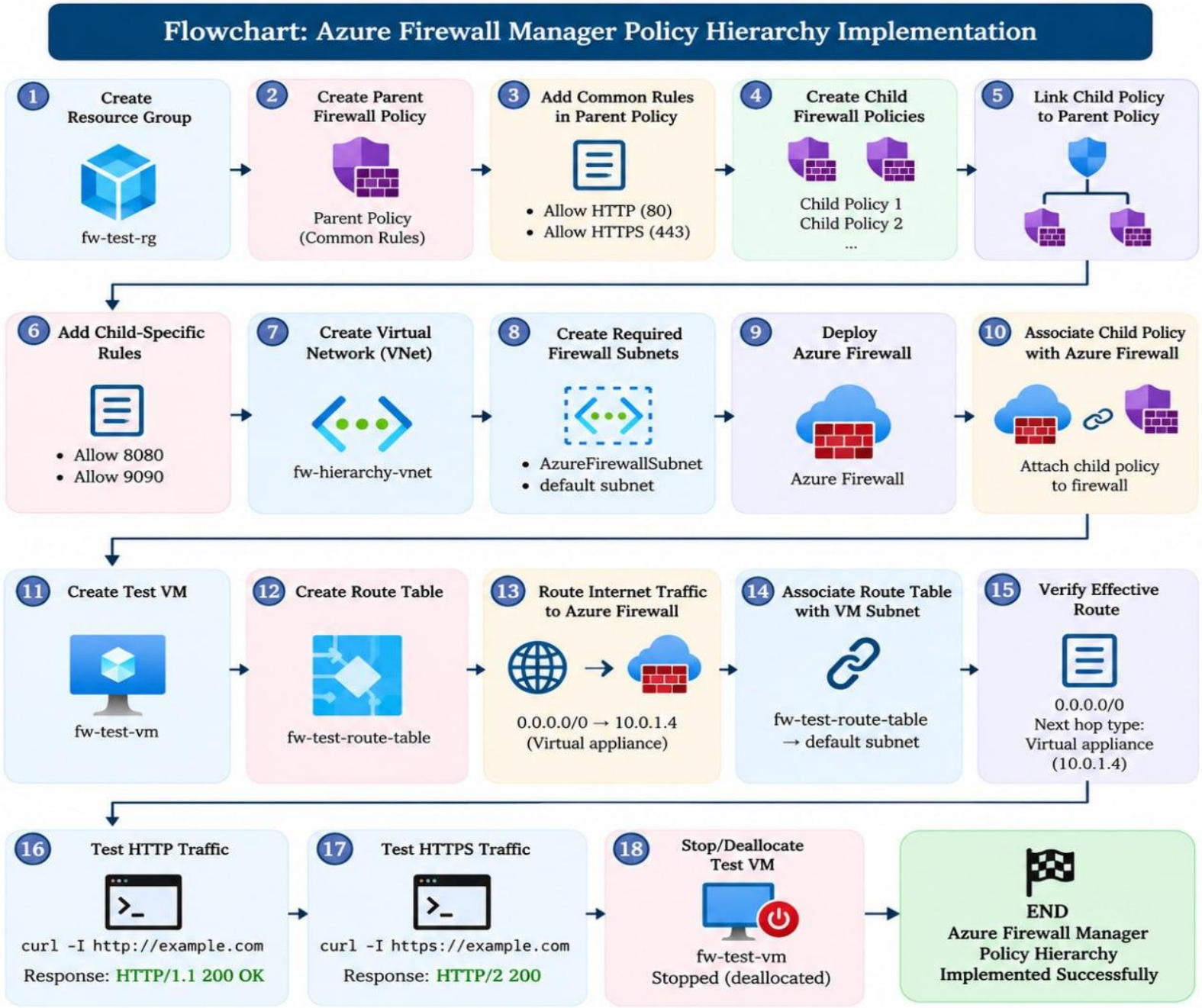
Step 17: HTTPS Traffic Test



Step 18 — Stop/Deallocate Test VM



FlowChart


Flowchart: Azure Firewall Manager Policy Hierarchy Implementation
1 Create Resource Group
fw-test-rg
2 Create Parent Firewall Policy
Parent Policy (Common Rules)
3 Add Common Rules in Parent Policy
Allow HTTP (80)
Allow HTTPS (443)
4 Create Child Firewall Policies
Child Policy 1
Child Policy 2
...
5 Link Child Policy to Parent Policy
6 Add Child-Specific Rules
Allow 8080
Allow 9090
7 Create Virtual Network (VNet)
fw-hierarchy-vnet
8 Create Required Firewall Subnets
AzureFirewallSubnet
default subnet
9 Deploy Azure Firewall
Azure Firewall
10 Associate Child Policy with Azure Firewall
Attach child policy to firewall
11 Create Test VM
fw-test-vm
12 Create Route Table
fw-test-route-table
13 Route Internet Traffic to Azure Firewall
0.0.0.0/0 → 10.0.1.4 (Virtual appliance)
14 Associate Route Table with VM Subnet
fw-test-route-table → default subnet
15 Verify Effective Route
0.0.0.0/0
Next hop type: Virtual appliance (10.0.1.4)
16 Test HTTP Traffic
curl -I http://example.com
Response: HTTP/1.1 200 OK
17 Test HTTPS Traffic
curl -I https://example.com
Response: HTTP/2 200
18 Stop/Deallocate Test VM
fw-test-vm
Stopped (deallocated)
END
Azure Firewall Manager Policy Hierarchy Implemented Successfully

Conclusion

Azure Firewall Manager provides a centralized way to manage firewall policies.

A **parent policy** was created for common network rules.

Child policies inherited the parent rules and added application-specific rules.

Azure Firewall was deployed and associated with the child policy.

A test VM and route table were configured to send Internet traffic through the firewall.

HTTP and HTTPS traffic were successfully tested.

The test VM was deallocated after testing to avoid unnecessary resource usage.